Melanoma Image Classification through CNN Image Analysis

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January 19, 2024

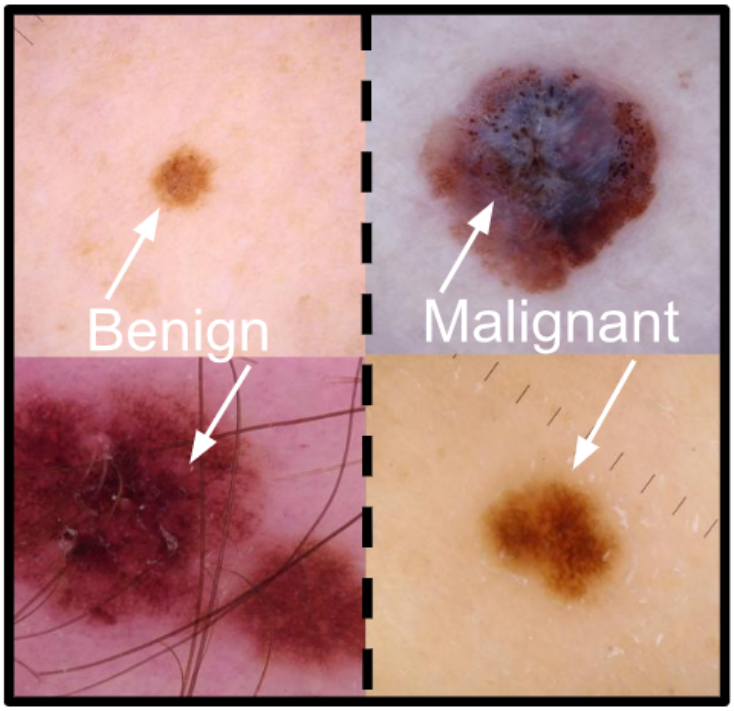

Figure 1: Images of 2 benign (healthy) and 2 malignant (cancerous).

Course 02461: Introduktion til Intelligente Systemer 2023-2024
Technical University of Denmark

1 Abstract

The rapid development of AI systems in the field of dermatology can assist medical personnel in early detection of malignant skin lesions, which is crucial for successfully curing patients. Our goal with this research project was to create a CNN with PyTorch able to correctly classify skin lesions. The CNN was trained on approx. 50000 images of skin lesions, either malignant (cancerous) or benign (healthy). The results show that the CNN's overall accuracy is 92.9%, while its specificity score is 97.9%. However, the CNN's "Recall" metric of 61.5% suggests that although the CNN excelled in identifying benign skin lesions, it had limitations with correctly identifying the malignant skin lesions.

Chapter	Main author
Introduction	Selma, Gabriel, Elias
Methodology	Selma, Gabriel
Results	Gabriel, Elias
Discussion	Elias, Selma

2 Introduction

When it comes to diagnosing skin cancer, AI will likely revolutionize both the technology and the accessibility of medical tools in the near future. One of the most well-known fields of AI, image recognition, seems to be taking the lead when it comes to assisting dermatologists around the world, as its ability to recognize patterns in scannings like images of skin lesions rival the abilities of specialists. In particular, CNNs (Convolutional Neural Networks), have been found incredibly efficient, as they can convolute medical images while still keeping the necessary features for them to be interpretable for the AI system. While this technology is truly profound, there are some things to be wary of when training and deploying this technology such as data biases and the consequences of false predictions. While a false positive might only result in an unnecessary health checkup, the consequence of the CNN predicting a false negative could lead to someone not getting cancer treatment in time. In this study we wish to build and train a CNN and maximize its prediction accuracy on a large dataset of skin lesion images, while also examining false prediction measurements like recall, specificity, and F1-score. We seek to investigate whether our CNN will outperform dermatologists in their initial assesment of a skin lesion, thus showing the usability of image recognition in skin cancer diagnostics. We also seek to understand if the model might be worse at predicting one of the two classes and why this happens.

3 Methodology

3.1 Dataset and Preprocessing

For this project, 3 datasets each containing images of different skin lesions were merged into one in order to increase the representative quality of the true population. Each of the images were tested and confirmed by medical professionals to be in one of two classes: either benign (healthy) or malignant (cancerous), which served as the "ground truth" and labels for the dataset. These labels were either given in a metafile or as sorted folder structures, so the first data preprocessing step was to use a python script[1] to iterate over each line in the metafiles and sort them. The first dataset, which was from the website Kaggle[2], contained 10000 images with a pixel size of 300x300 pixels. The second [3] and third[4] datasets contained approx. 10000 and 33000 images respectively, and they varied greatly in pixel size. The image size of 300x300 pixels was chosen as the standard for the combined dataset, as it was seemingly a good balance between a high processing effectiveness and having identifiable features. The combined dataset contained 46600 benign and 7100 malignant images of skin lesions, which can be considered quite an imbalance (more on this in the discussion chapter). Finally, the dataset was split up into 80%/10%/10% for a training set, a validation set and a test set respectively. The decision to partition the dataset into test and validation sets aimed to mitigate the risk of manual overfitting from hyperparameter

[1] https://github.com/edlunoe/Group-10-Project-Folder
[2] JAVID, M. H.
[3] Tschandl, Philipp, 2018
[4] Rotemberg, et al.

tuning later in the process.

3.2 Model Architecture

Many different combinations of hyperparameters and layers were tried when deciding on how to build the CNN in the Python library PyTorch. In the end, a fairly standard CNN with 2 convolutional layers was chosen in the pursuit of finding optimal hyperparameters. Each convolutional layer consisted of a convolution with a ReLU activation function that introduced non-linearity to the CNN by setting all negative values in the emerging feature maps to 0, while positive values stayed unchanged. Each Convolution and ReLU were followed up by a Max Pooling layer that shrunk the feature maps to half their size. Following the feature extraction of the two convolutional layers, a flattening layer converted the input into a one dimensional tensor, which was required for the subsequent two fully connected layers (see figure 2). The first linear layer transformed the $32 \times 73 \times 73$ long one-dimensional input tensor into 128 output features using linear transformation. This was followed up by yet another ReLU activation function. Next, the second fully connected layer took 128 input features and gave 1 output feature. Lastly, a sigmoid activation function was used for binary classification, implicitly predicting the positive class (malignant) if over a certain threshold of 0.5, and otherwise it would predict negative (benign).

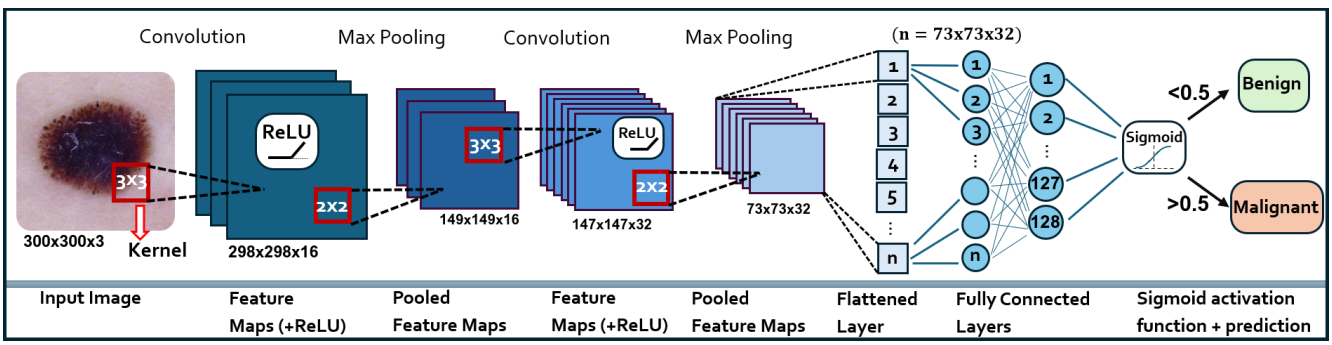

Figure 2: Illustration of architecture of our CNN visualized with specific image sizes and kernel sizes included.

It is worth noting that implementing dropout layers before and after pooling layers was tested as a regularization technique to combat any overfitting by the model. Further testing showed that a dropout of zero returned the highest training and validation accuracy as well as the lowest loss, after numerous iterations of hyperparameter tuning. This was likely due to the relatively small size of the CNN's architecture making regularization redundant (see figure 3).

3.3 Training Process

After deciding on an appropriate CNN architecture, a problem emerged from the pilot stages of training that had to be solved. Having a relatively large dataset resulted in slow training speed, which greatly reduced the timeframe for experimenting with different hyperparameters, etc. This was solved by switching from CPU to cuda (GPU), giving the CNN a 15x decrease in time spent training. Furthermore, a batch size of 32 and the function arguments "pin_memory = True" and

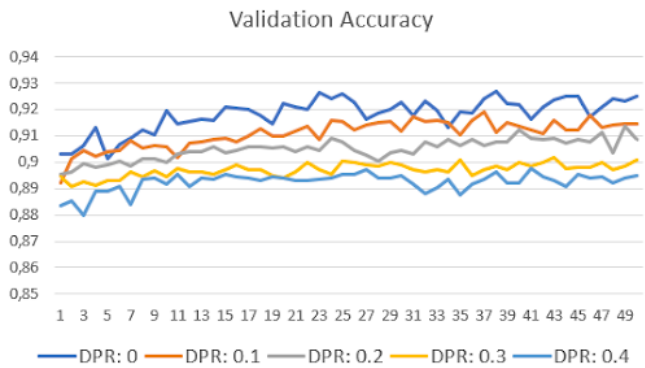

Figure 3: Validation accuracy over 50 epochs with dropout values ranging from 0 to 0.4.

"numworkers = 2" in the data loader gave the CNN an increase in training efficiency. Adam (short for Adaptive Moment Estimation) was chosen as the optimization algorithm for the CNN based on standard practice and an acceptable performance in the pilot studies. Another standard choice was made in regards to the error function, as the Binary Cross-Entropy loss function was implemented for the training loop. Each instance of training lasted for 100 epochs, after which the epoch showing the highest validation accuracy was saved and loaded into a test script. The CNN model was then applied to the test dataset with unseen skin lesion images. The results were plotted in a confusion matrix showing the final distribution of True Negatives (TN), False Negatives (FN), True Positives (TP) and False Positives (FP). Negative being benign, positive being malignant and true/false being correct and incorrect classifications respectively. Since the combined dataset was relatively unbalanced classwise, it was decided to also train the CNN on the smaller, but more balanced dataset from the Kaggle Database[5] containing approx. 5000 images of each class. This was done to see how the class distribution in the dataset would affect overall performance of the CNN.

4 Results

The top diagram of figure 4 shows the confusion matrix of the finalized CNN model when tested and trained on the combined dataset. The bottom diagram of figure 4 represents the confusion matrix for the CNN trained and tested on only the balanced Kaggle dataset[6]. While overall test accuracy is an important metric in evaluating the model's performance, it is essential to use other evaluation metrics that take the imbalance of the dataset into account. The metrics chosen to give a more in depth understanding of the results are Specificity (true negative rate), Recall (true positive rate/sensitivity) and a F1-

[5]JAVID, M. H.
[6]JAVID, M. H.

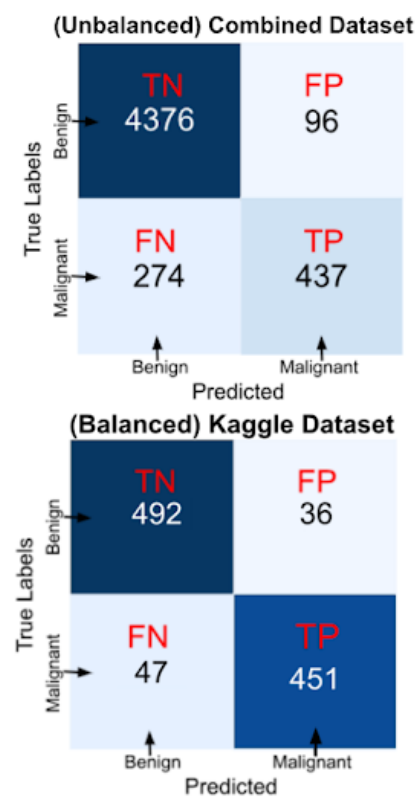

Figure 4: Confusion matrices

score accompanied by a 99% CI for each metric [7]. These different metrics can be calculated as follows:

$$\text{Accuracy} = \frac{TP + TN}{TN + FN + TP + FP}$$
$$\text{Specificity} = \frac{TN}{TN + FP}$$
$$\text{Recall} = \frac{TP}{TP + FN}$$
$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Table 1: Comparison of Datasets

	(Unbalanced) Combined Dataset			(Balanced) Kaggle Dataset		
Measurement	Mean	99% CI	p value	Mean	99% CI	p value
Test accuracy	0.929	[0.919, 0.938]	$< 2.2e-16$	0.918	[0.894, 0.939]	$< 2.2e-16$
Specificity	0.979	[0.972, 0.983]	$< 2.2e-16$	0.930	[0.896, 0.956]	$< 2.2e-16$
Recall	0.615	[0.566, 0.661]	$< 2.2e-16$	0.906	[0.867, 0.936]	$< 2.2e-16$
F1-score	0.703	[0.699, 0.706]	$< 2.2e-16$	0.915	[0.909, 0.921]	$< 2.2e-16$

Table 1: All of the mentioned metrics are given in the table above, both for the combined dataset and the balanced Kaggle dataset. The test accuracy is the overall ratio of correct prediction over the total number of predictions. Specificity is the proportion of correct negative cases predicted compared to all negatives predicted. Recall is the proportion of correct positive cases predicted compared to all positive predictions. F1-score is a score that takes the recall and precision into account, where precision being a measure for the ratio of true positive predictions to the total number of positive predictions made. Furthermore, a weighting has been implemented in the F1-score so a higher penalty is warranted for false negatives.

5 Discussion

When only taking the overall test accuracy of 92.9% into account, the CNN seems to be performing relatively well on the combined dataset. However, the F1-score and especially the Recall is suboptimal in regards to potential usage as a diagnostics medical tool. On the contrary, the balanced Kaggle dataset gives the CNN a lower test accuracy of 91.8%, but the Recall and F-score of 90.6% and 91.5% respectively are significantly better. However, one may argue that the larger dataset with 50000 images better represents the true population of skin lesions, as the balanced Kaggle dataset only has 10000 images. By training the CNN on images from the three different datasets it makes the model more robust, and thus it might perform better when faced with new images. Therefore, only the CNN trained on the combined dataset will be discussed further.

[7] See R-document: https://github.com/edlunoe/Group-10-Project-Folder

With the large dataset, the CNN isn't able to outperform specialist dermatologists, whose Specificity and Recall has been measured in a "Humans against Machines" contest of classifying skin lesions: "In level-I, dermatologists achieved a mean (±standard deviation) [recall] and specificity for lesion classification of 86.6% (±9.3%) and 71.3% (±11.2%),"[8]. Man against machine: diagnostic performance of a deep learning convolutional neural network for dermoscopic melanoma recognition in comparison to 58 dermatologists - ScienceDirect In comparison to the dermatologists, the CNN has a much lower Recall but a higher Specificity. This might be due to the fact that the doctors generally want to be on the safe side with an odd skin lesion (one would rather perform an excessive checkup than risk a patient's life). This idea of being extra safe can also be implemented in the CNN's architecture, for instance by changing the threshold of the CNN output, thus giving more positive predictions. However, this doesn't fix the issue that the CNN gives subpar outputs, which might be because of the imbalance in the dataset. One option to handle this would be weighting the BCE loss function to increase the cost of a false negative prediction, as these are the most dangerous false predictions. While this concept is reasonable in theory, it did not turn out very well with our CNN, as a weighted BCE loss function seemingly decreased the overall accuracy while not eliminating any false negatives. Another reason for the big difference in Recall and Specificity is likely due to the fact that the CNN is limited by only being able to analyze three out of the five characteristics of the ABCDEs of melanoma detection[9]. These five characteristics are asymmetry, odd borders, color, diameter, and evolution over time. The CNN's inability to assess the mole's diameter and evolution from a single image inhibits it to detect two of the crucial factors in cancer identification. Because of this, the CNN is unable to detect if a benign looking skin lesion has experienced rapid growth as a result of cancer. To optimize the CNN's performance it would therefore be necessary to have additional information such as absolute diameter size and evolution over time. This would give the CNN the best prerequisites to identify a cancerous skin lesion. To summarize, there are many different ways to improve the CNN, thereby making it more applicable in real life for skin cancer diagnostics. While it can be argued that the model trained on the balanced Kaggle dataset yields more preferable results with a recall of 90.6%, we firmly support our decision to focus on the imbalanced and larger dataset with a lower recall of 61.5%. This CNN provides less impressive but more realistic results, being tested on images from all three datasets.

[8] H. Haenssle et al.
[9] Mogensen, Mette, et al.

6 Appendix

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8. Github Folder with all codes snippets and R-documents:

   https://github.com/edlunoe/Group-10-Project-Folder